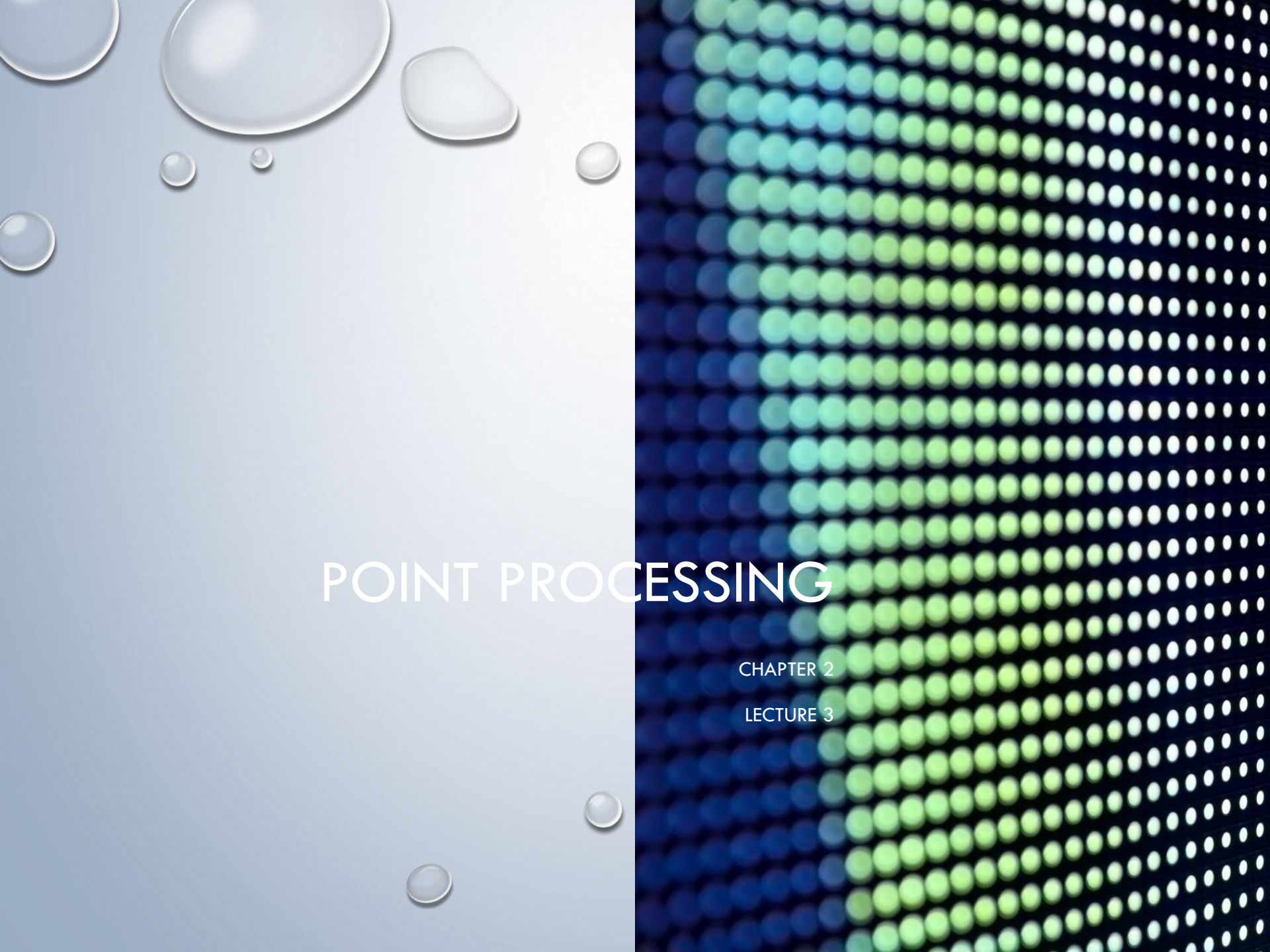
IMAGE PROCESSING

REFERENCE BOOK:

AN INTRODUCTION TO DIGITAL IMAGE PROCESSING WITH MATLAB

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POINT PROCESSING

CHAPTER 2

LECTURE 3

IMAGE PROCESSING OPERATIONS TYPES

Image processing operations may be divided into into three classes based on the information required to perform the transformation, they are:

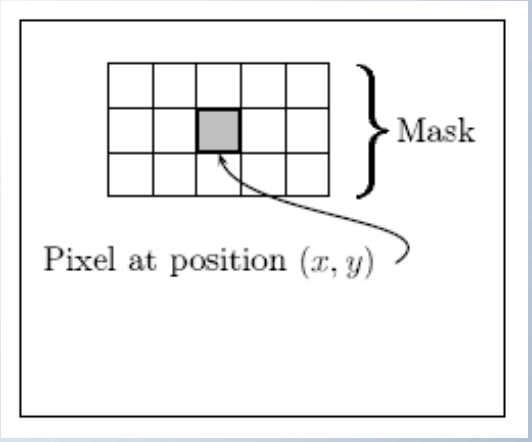
- Point Process
- Neighborhood process
- Transform

POINT OPERATIONS

- A PIXEL'S GREY VALUE IS CHANGED WITHOUT ANY KNOWLEDGE OF ITS SURROUNDS.
- IN A DIGITAL IMAGE, POINT = PIXEL.
- POINT PROCESSING TRANSFORMS A PIXEL’S VALUE AS FUNCTION OF ITS VALUE ALONE;
- IT DOES NOT DEPEND ON THE VALUES OF THE PIXEL’S NEIGHBORS.

NEIGHBORHOOD PROCESSING.

- TO CHANGE THE GREY LEVEL OF A GIVEN PIXEL WE NEED ONLY KNOW THE VALUE OF THE GREY LEVELS IN A SMALL NEIGHBORHOOD OF PIXELS AROUND THE GIVEN PIXEL.



TRANSFORMS.

- A TRANSFORM REPRESENTS THE PIXEL VALUES IN SOME OTHER, BUT EQUIVALENT FORM. TRANSFORMS ALLOW FOR SOME VERY EFFICIENT AND POWERFUL ALGORITHMS. WE MAY CONSIDER THAT IN USING A TRANSFORM, THE ENTIRE IMAGE IS PROCESSED AS A SINGLE LARGE BLOCK.

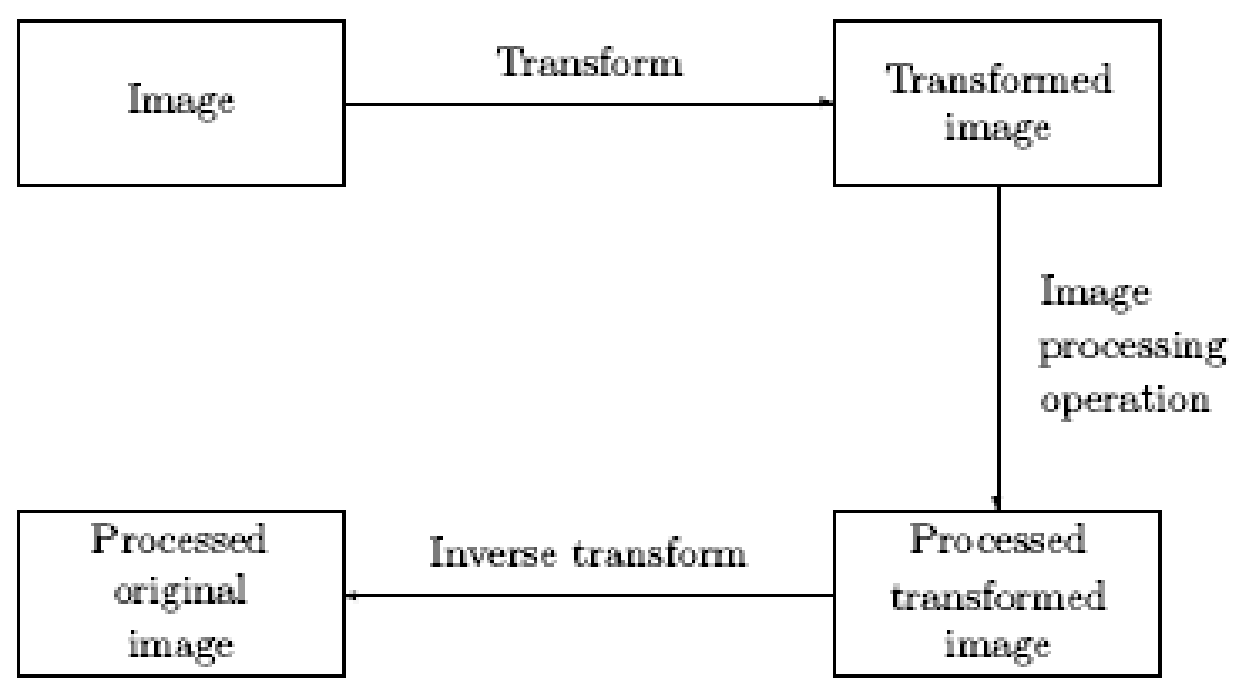


Figure 2.1: Schema for transform processing

POINT PROCESS

- MORE SIMPLE
- THEY ARE ESPECIALLY USEFUL IN IMAGE PRE-PROCESSING, WHERE AN IMAGE IS REQUIRED TO BE MODIFIED BEFORE THE MAIN JOB IS ATTEMPTED.
- THE MOST PRIMITIVE, YET ESSENTIAL, IMAGE PROCESSING OPERATIONS. - INTENSITY TRANSFORMATIONS THAT CONVERT AN OLD PIXEL INTO A NEW PIXEL BASED ON SOME PREDEFINED FUNCTION.
- - USED PRIMARILY FOR CONTRAST ENHANCEMENT.

POINT PROCESSING OF IMAGES

- Brightness and contrast adjustment
- Gamma correction
- Histogram equalization
- Histogram matching
- Color correction.

2.2 Arithmetic operations

These operations act by applying a simple function

$$y = f(x)$$

Simple functions include adding or subtract a constant value to each pixel:

$$y = x \pm C$$

Or Multiplying a constant value to each pixel:

$$y = Cx.$$

In order to ensure that the output value ranges from 0 to 255 you have to:
Round the output values to an integer values
Clipping the output value to ranges from 0 to 255

$$y \leftarrow \begin{cases} 255 & \text{if } y > 255, \\ 0 & \text{if } y < 0. \end{cases}$$

Adding and subtracting a constant

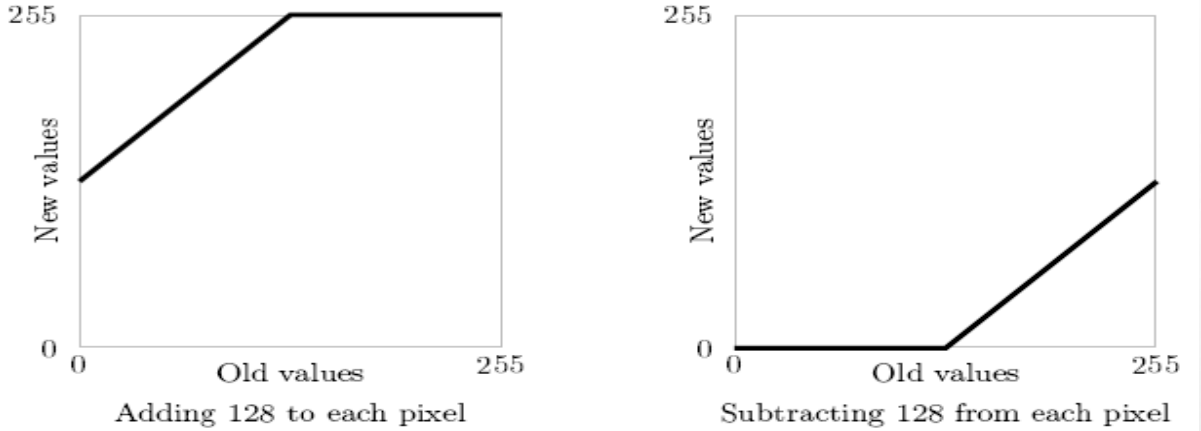
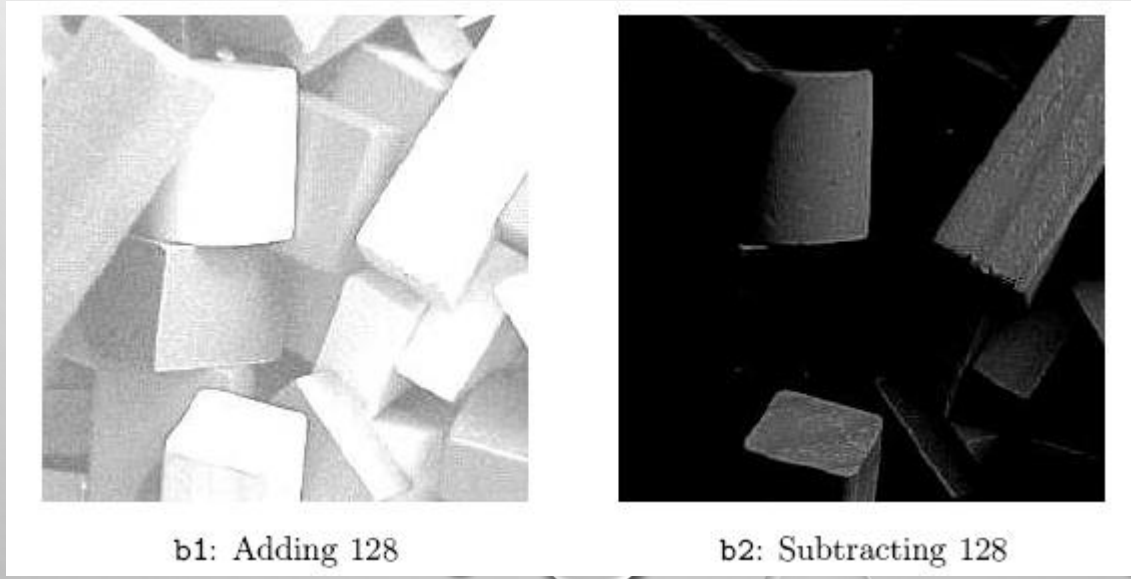


Figure 2.2: Adding and subtracting a constant

We can test this on the blocks image blocks.tif



b1: Adding 128

b2: Subtracting 128

Adding and subtracting a constant

We can test this on the blocks image blocks.tif

```
>> b=imread('blocks.tif');
>> whos b
  Name      Size            Bytes  Class

  b       256x256           65536  uint8 array
```

```
>> b1=b+128
??? Error using ==> +
Function '+' not defined for variables of class 'uint8'.
```

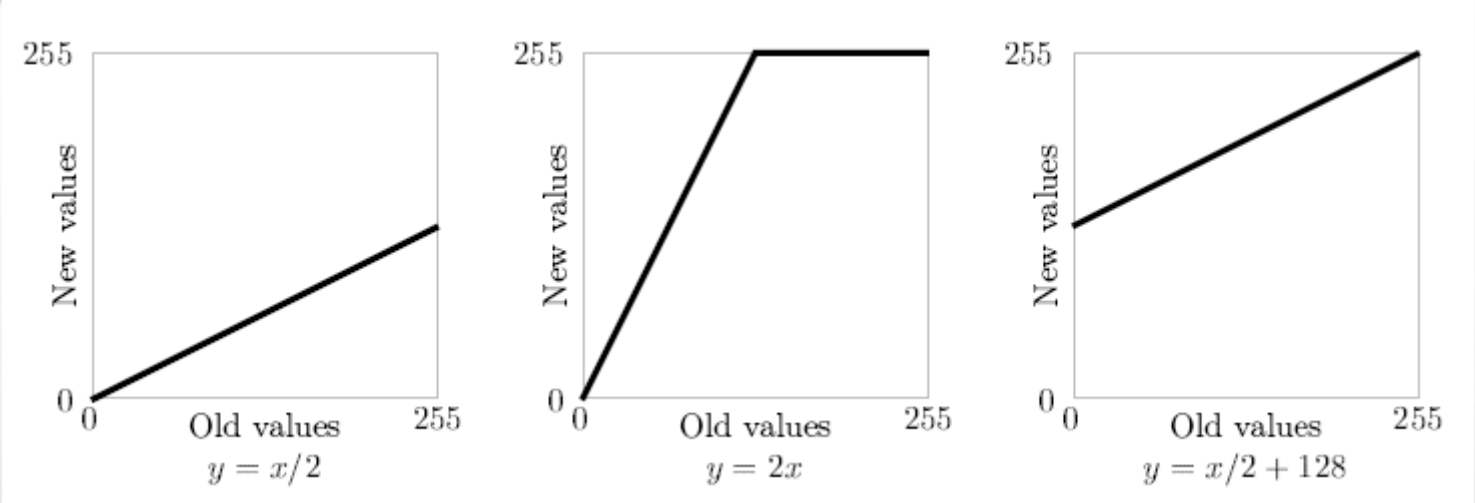
```
>> b1=uint8(double(b)+128);
```

```
>> b1=imadd(b,128);
```

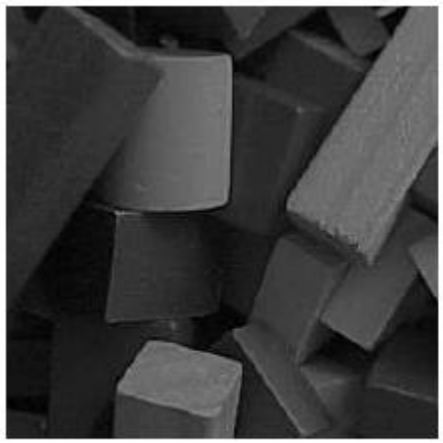
```
>> b2=imsubtract(b,128);
```

```
>> imshow(b1),figure,imshow(b2)
```

Using multiplication and division



$y = x/2$	`b3=immultiply(b,0.5);` or `b3=imdivide(b,2)`
$y = 2x$	`b4=immultiply(b,2);`
$y = x/2 + 128$	`b5=imadd(immultiply(b,0.5),128);` or `b5=imadd(imdivide(b,2),128);`



b3: $y = x/2$



b4: $y = 2x$



b5: $y = x/2 + 128$

Complements

```
>> bc=imcomplement(b);
>> imshow(bc)
```

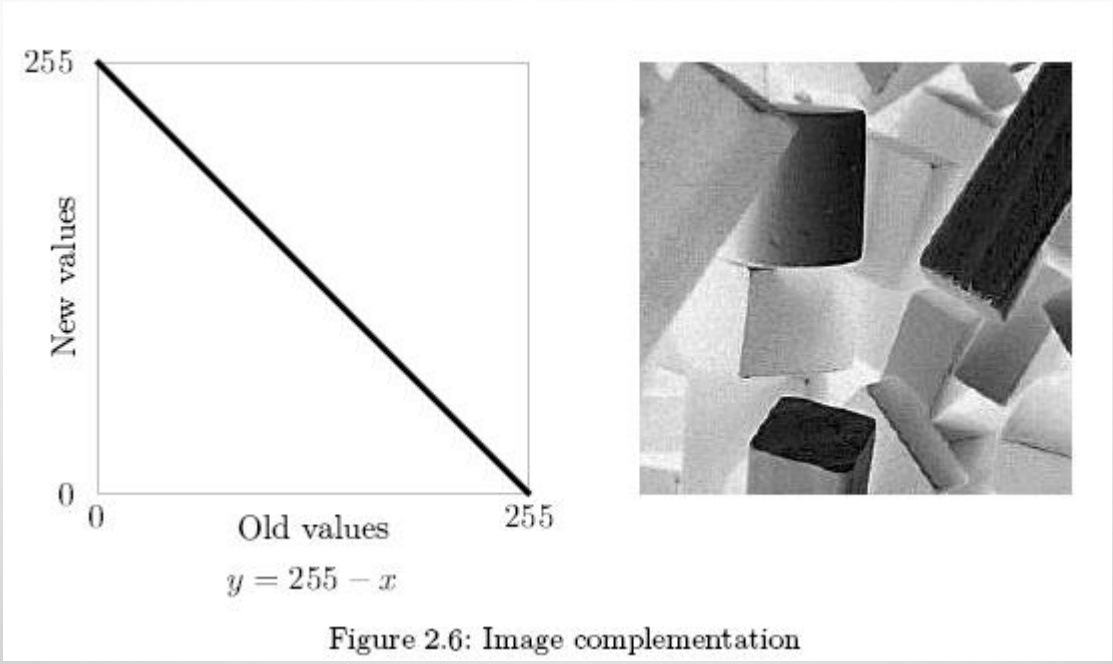
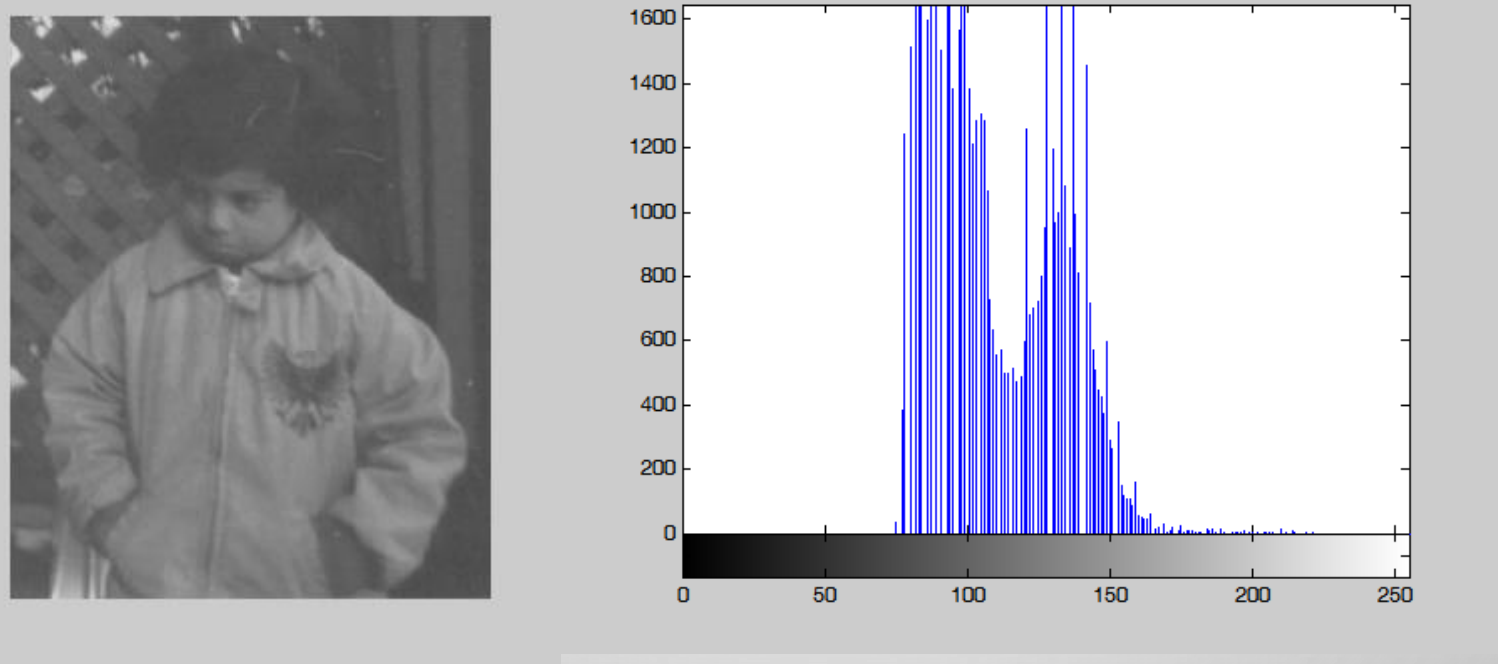


Figure 2.6: Image complementation

Interesting special efects can be obtained by complementing only part of the image; for example by taking the complement of pixels of grey value 128 or less, and leaving other pixels untouched. Or we could take the complement of pixels which are 128 or greater, and leave other pixels untouched.The eect of these functions is called solarization.

Histograms

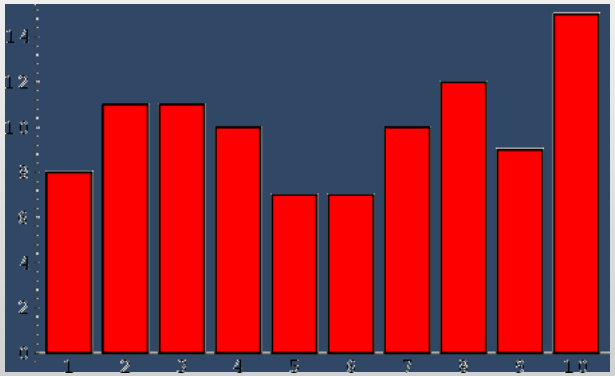
Given a grayscale image, its histogram consists of the histogram of its grey levels; that is, a graph indicating the number of times each grey level occurs in the image



The image pout.tif and its histogram

WHAT IS A HISTOGRAM?

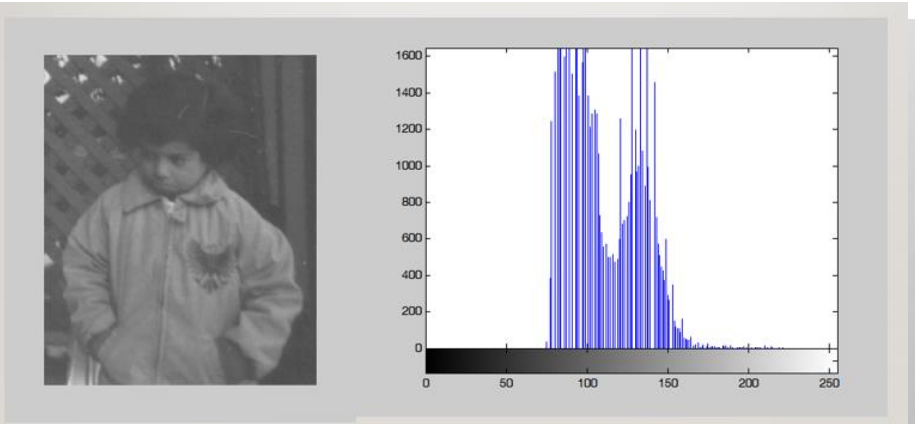
- A GRAPH INDICATING THE NUMBER OF TIMES EACH GRAY LEVEL OCCURS IN THE IMAGE, I.E. **FREQUENCY** OF THE **BRIGHTNESS VALUE** IN THE IMAGE
- THE HISTOGRAM OF AN IMAGE WITH *L* GRAY LEVELS IS REPRESENTED BY A ONE-DIMENSIONAL ARRAY WITH *L* ELEMENTS.
- ALGORITHM:
  - ASSIGN ZERO VALUES TO ALL
  - ELEMENTS OF THE ARRAY HF;
  - FOR ALL PIXELS (X,Y) OF THE
  - IMAGE F, INCREMENT
  - HF[F(X,Y)] BY 1.



HISTOGRAM STRETCHING (CONTRAST STRETCHING)

- SUPPOSE WE HAVE AN IMAGE WITH THE HISTOGRAM SHOWN IN FIGURE , ASSOCIATED WITH A TABLE OF THE NUMBERS
- WE CAN STRETCH THE GREY LEVELS IN THE CENTRE OF THE RANGE OUT BY APPLYING THE PIECEWISE LINEAR FUNCTION SHOWN AT THE RIGHT IN FIGURE 2.9. THIS FUNCTION HAS THE EFFECT OF STRETCHING THE GREY LEVELS – TO GREY

Grey level i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n_i	15	0	0	0	0	70	110	45	70	35	0	0	0	0	0	15



The image pout.tif and its histogram

HISTOGRAM STRETCHING(CONTRAST STRETCHING)

- WHERE I IS THE ORIGINAL GREY LEVEL AND J ITS RESULT AFTER THE TRANSFORMATION. GREY LEVELS OUTSIDE THIS RANGE ARE EITHER LEFT ALONE (AS IN THIS CASE) OR TRANSFORMED ACCORDING TO THE LINEAR FUNCTIONS AT THE ENDS OF THE GRAPH ABOVE. THIS YIELDS:

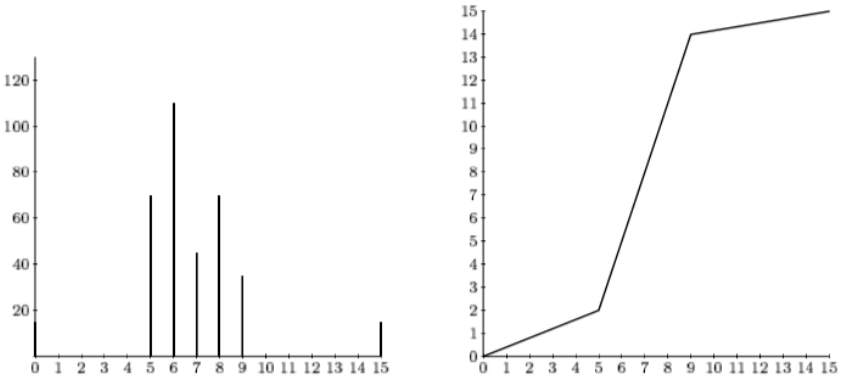
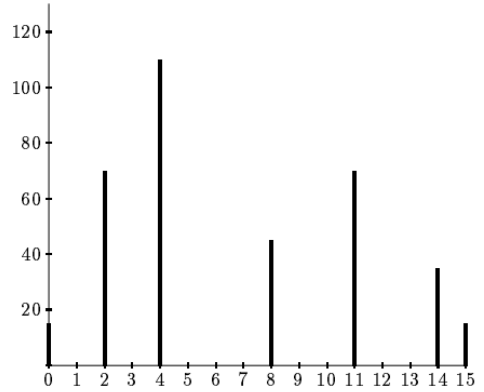


Figure 2.9: A histogram of a poorly contrasted image, and a stretching function

Mapping function

$$j = \frac{14-2}{9-5}(i-5)+2$$

THE RESULT WILL BE:



i	5	6	7	8	9
j	2	5	8	11	14

USE OF IMADJUST

- IMADJUST(IM,[A,B],[C,D])

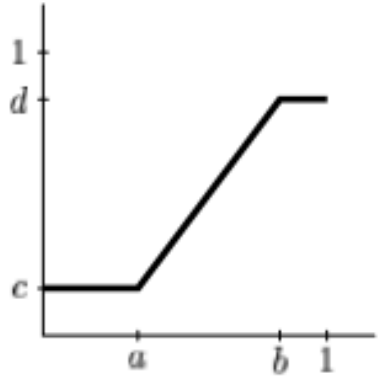


Figure 2.10: The stretching function given by `imadjust`

GAMMA CORRECTION

The function used is a slight variation on the standard line between two points:

$$y = \left(\frac{x-a}{b-a}\right)^{\gamma}(d-c)+c.$$

Use of the gamma value alone can be enough to substantially change the appearance of the image
For example:

```
>> t=imread('tire.tif');
>> th=imadjust(t,[],[],0.5);
>> imshow(t),figure,imshow(th)
```

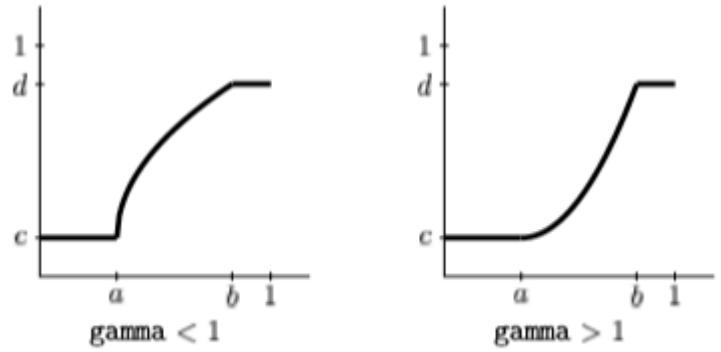


Figure 2.11: The imadjust function with gamma not equal to 1

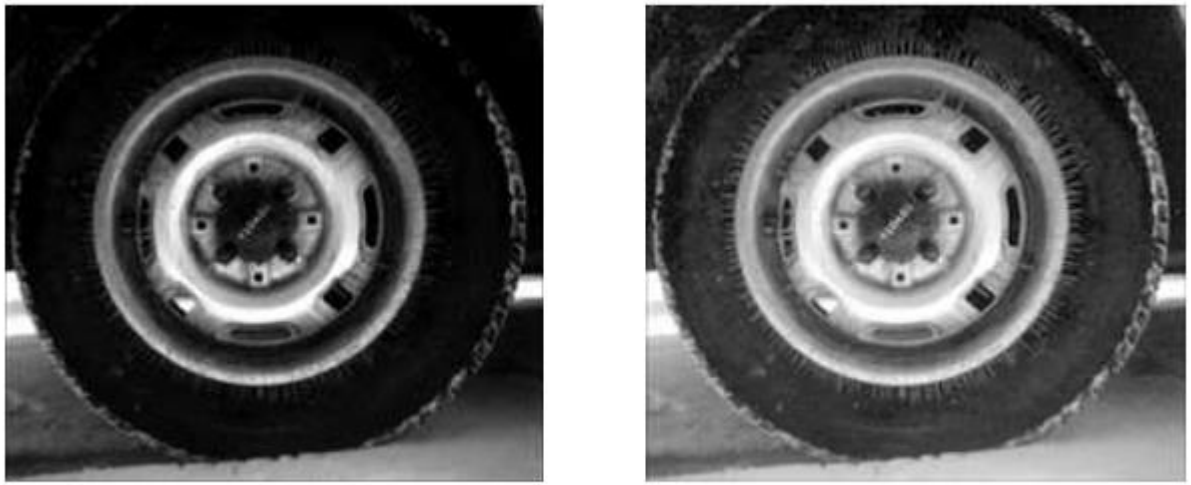
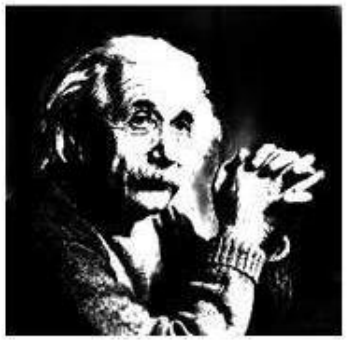


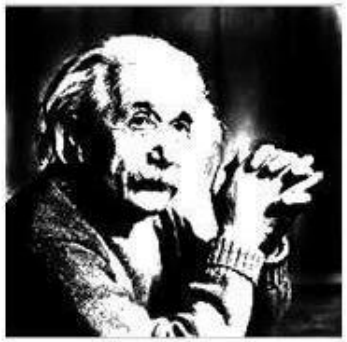
Figure 2.12: The tire image and after adjustment with the gamma value

EXAMPLES OF GAMM CORRECTION

Gamma = 10



Gamma = 8



Gamma = 6



A PIECEWISE LINEAR STRETCHING FUNCTION

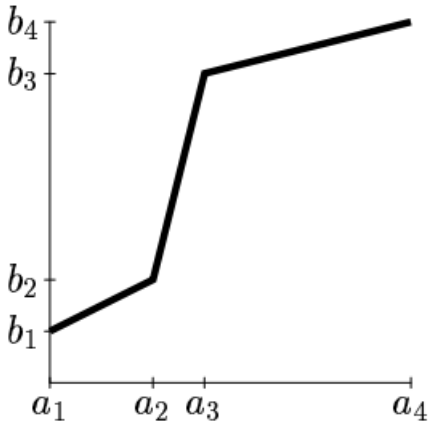


Figure 2.14: A piecewise linear stretching function

$$y = \frac{b_{i+1} - b_i}{a_{i+1} - a_i}(x - a_i) + b_i$$

HISTOGRAM EQUALIZATION

- THE TROUBLE WITH ANY OF THE ABOVE METHODS OF HISTOGRAM STRETCHING IS THAT THEY REQUIRE USER INPUT. SOMETIMES A BETTER APPROACH IS PROVIDED BY HISTOGRAM EQUALIZATION, WHICH IS AN ENTIRELY AUTOMATIC PROCEDURE.
- THE IDEA IS TO CHANGE THE HISTOGRAM TO ONE WHICH IS UNIFORM; THAT IS THAT EVERY BAR ON THE HISTOGRAM IS OF THE SAME HEIGHT, OR IN OTHER WORDS THAT EACH GREY LEVEL IN THE IMAGE OCCURS WITH THE SAME FREQUENCY.
- IN PRACTICE THIS IS GENERALLY NOT POSSIBLE, ALTHOUGH AS WE SHALL SEE THE RESULT OF HISTOGRAM EQUALIZATION PROVIDES VERY GOOD RESULTS.

HISTOGRAM EQUALIZATION EXAMPLE

- SUPPOSE OUR IMAGE HAS L DIFFERENT GREY LEVELS 0,1,..,L-1.

Grey level i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n_i	15	0	0	0	0	0	0	0	0	70	110	45	80	40	0	0

Grey level i	n_i	Σn_i	$(1/24)\Sigma n_i$	Rounded value
0	15	15	0.63	1
1	0	15	0.63	1
2	0	15	0.63	1
3	0	15	0.63	1
4	0	15	0.63	1
5	0	15	0.63	1
6	0	15	0.63	1
7	0	15	0.63	1
8	0	15	0.63	1
9	70	85	3.65	4
10	110	195	8.13	8
11	45	240	10	10
12	80	320	13.33	13
13	40	360	15	15
14	0	360	15	15

Original grey level i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Final grey level j	1	1	1	1	1	1	1	1	1	4	8	10	13	15	15	15

FINAL RESULT OF HISTOGRAM EQUALIZATION

```
>> p=imread('pout.tif');
>> ph=histeq(p);
>> imshow(ph),figure,imhist(ph),axis tight
```



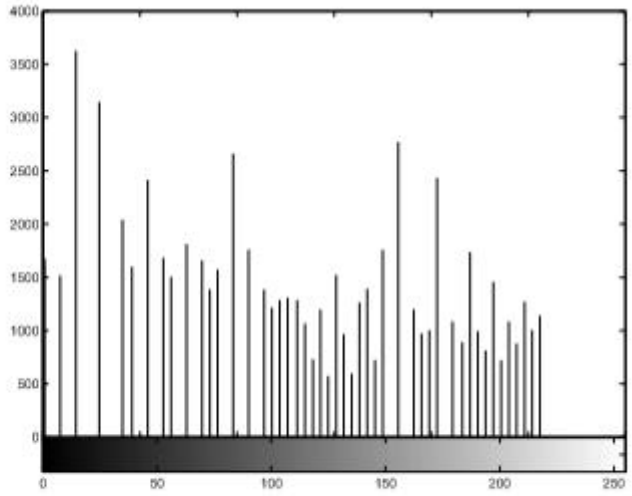


Figure 2.19: The histogram of figure 2.8 after equalization

EXAMPLE 2

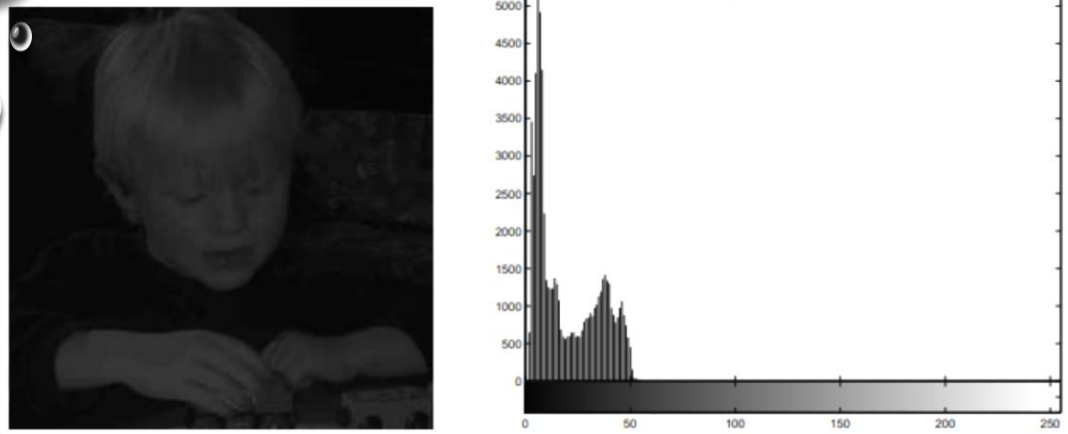


Figure 2.20: The darkened version of `engineer.tif` and its histogram

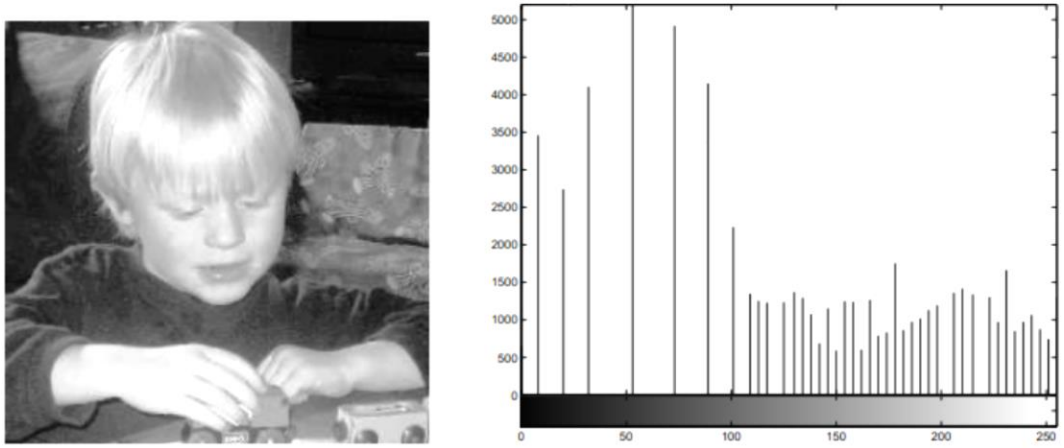


Figure 2.21: The image from 2.20 equalized and its histogram